LUMIA BASED STRAM CONTROLLER SETUP GUIDE

OVERVIEW:

WHAT THIS IS

THIS GUIDE COVERS SETTING UP A LUMI-CON CONTROLLER USING:

- ESP8266 (WI-FI + LUMIA + OPTIONAL TFT SCREEN)
- RASPBERRY PI PICO (SCANS THE 6×6 KEYPAD MATRIX)
- OPTIONAL: 1.8" ST7735 TFT SCREEN FOR CHAT/STATUS DISPLAY

PARTS REQUIRED

- ESP8266
- RASPBERRY PI PICO
- 36× 1N4148 DIODES
- 36× MECHANICAL SWITCHES
- 1.8" RGB TFT ST7735 (ONLY IF YOU WANT CHAT/SCREEN FEATURES)

LUMI-CON

FILES REQUIRED

ESP8266 FIRMWARE (INTEGRATED FIRMWARE):

- LUMI_CON_ESP_INTEGRATED_0_0_4

PICO FIRMWARE (6×6 SCANNER):

- 6X6_TEST_PICO

LUMIA PLUGIN (PICK ONE)

A) JUST CHAT (SCREEN ONLY)

- ESP8266-TFT-CHAT-BRIDGE

B) JUST 6×6 (BUTTONS ONLY)

- PLUGIN VERSION 5.2

C) BOTH (BUTTONS + SCREEN)

- PLUGIN VERSION 5.2

IMPORTANT: PICK YOUR PLUGIN NOW. IT CHANGES WHAT STEPS YOU NEED LATER.

LUMI-CON 6X6 MINI

LUMI-CON CHAT BRIDGE MINI

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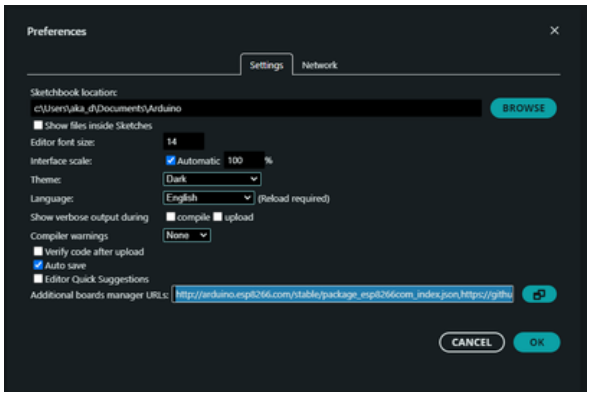
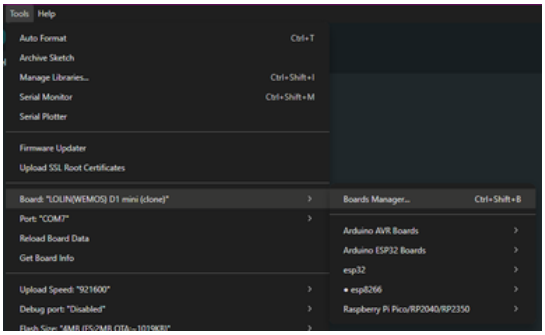
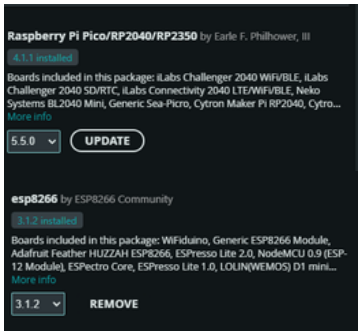
FIRMWARE FLASHING:



STEP 2



STEP 3



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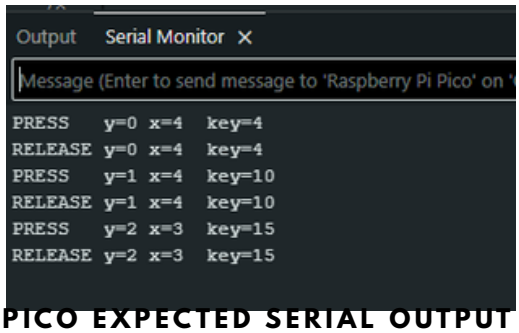

PICO EXPECTED SERIAL OUTPUT

STEP 1: INSTALL ARDUINO IDE
DOWNLOAD ARDUINO IDE FROM ARDUINO'S WEBSITE AND OPEN IT.

STEP 2: ADD BOARD MANAGER URLS
ARDUINO IDE → FILE → PREFERENCES
ADD THE URL(S) FROM YOUR "BOARD MANAGER URL" TEXT FILE INTO "ADDITIONAL BOARDS MANAGER URLS".

STEP 3: INSTALL BOARD PACKAGES
ARDUINO IDE → TOOLS → BOARD → BOARDS MANAGER
INSTALL:
- RASPBERRY PI PICO / RP2040 BOARD PACKAGE
- ESP8266 BOARD PACKAGE

STEP 4: FLASH THE PICO (6×6 SCANNER)
1. PLUG THE PICO INTO YOUR PC WITH USB
2. OPEN THE 6X6_TEST_PICO SKETCH IN ARDUINO IDE
3. CLICK UPLOAD
4. OPEN SERIAL MONITOR
5. PRESS KEYS ON YOUR KEYPAD
6. YOU SHOULD SEE CONFIRMATION MESSAGES APPEAR IN SERIAL MONITOR.

STEP 5: FLASH THE ESP8266 (WI-FI + LUMIA + OPTIONAL TFT)
IMPORTANT: DURING ESP FLASHING, DISCONNECT THE ESP RX PIN FROM THE PICO.
(IF RX IS CONNECTED, UPLOADING CAN FAIL.)
1. PLUG ESP8266 INTO USB
2. OPEN LUMI_CON_ESP_INTEGRATED IN ARDUINO IDE
3. CLICK UPLOAD
4. WHEN THE ESP SUCCESSFULLY RECEIVES A BUTTON PRESS, THE ESP'S ON-BOARD LED SHOULD FLASH.

AFTER FLASHING IS COMPLETE: RECONNECT THE PICO → ESP RX WIRE IF YOU ARE USING THE 6×6 BUTTONS.

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PLUGIN BUILD AND INSTALL

STEP 1: CREATE A PLUGIN BUILD FOLDER

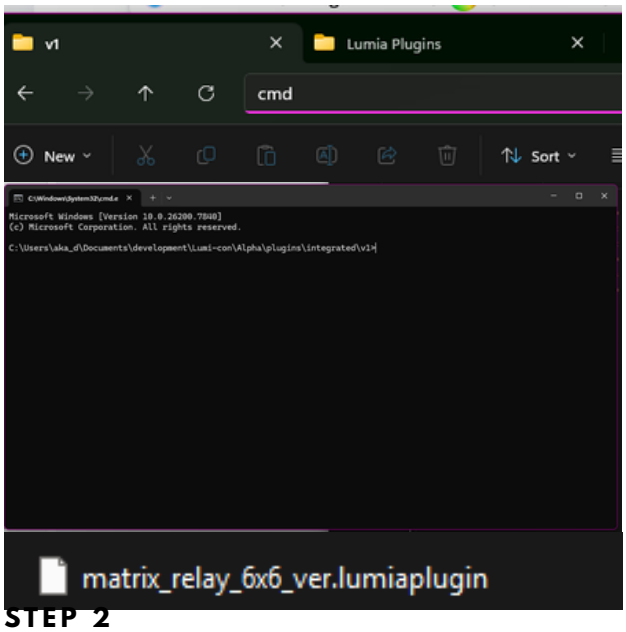
1. MAKE A NEW FOLDER ANYWHERE (DESKTOP IS FINE)
2. NAME IT SOMETHING OBVIOUS (EXAMPLE: LUMI_PLUGIN_BUILD)
3. COPY THE PLUGIN FILES YOU WANT TO USE INTO THIS FOLDER

STEP 2: OPEN COMMAND PROMPT IN THAT FOLDER

IN FILE EXPLORER, CLICK THE ADDRESS BAR, TYPE:

CMD

PRESS ENTER. A COMMAND PROMPT OPENS IN THAT FOLDER.



STEP 2

STEP 3: BUILD THE PLUGIN

RUN THESE COMMANDS:

```
NPM INSTALL
NPX LUMIA-PLUGIN VALIDATE .
NPX LUMIA-PLUGIN BUILD . --OUT .
\LUMI_CON_BRIDGE_INTEGRATED_V5_2.LUMIAPLUGIN
```

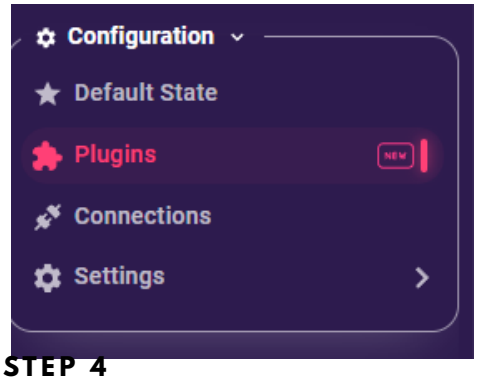
RENAME "LUMI_CON_BRIDGE_INTEGRATED_V5_2" TO WHATEVER YOU WANT THE OUTPUT FILE TO BE CALLED.

STEP 4: INSTALL INTO LUMIA

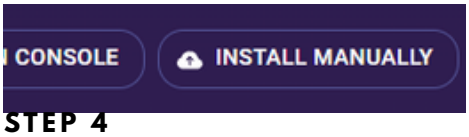
IN LUMIA:

CONFIGURATION → PLUGINS → INSTALLED → INSTALL MANUALLY

DRAG IN (OR SELECT) THE .LUMIAPLUGIN FILE YOU JUST CREATED.



STEP 4



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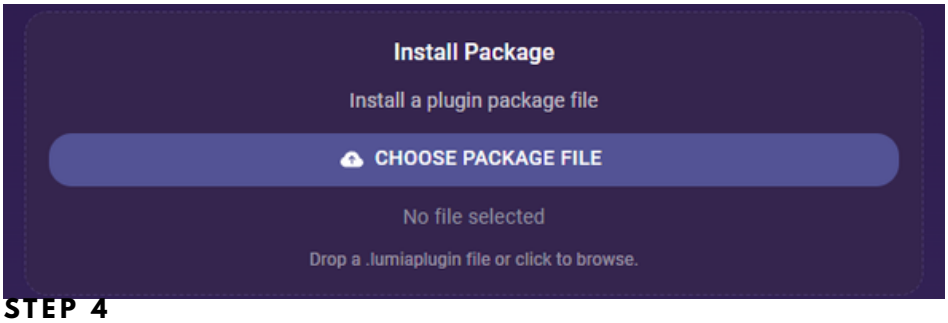

STEP 4

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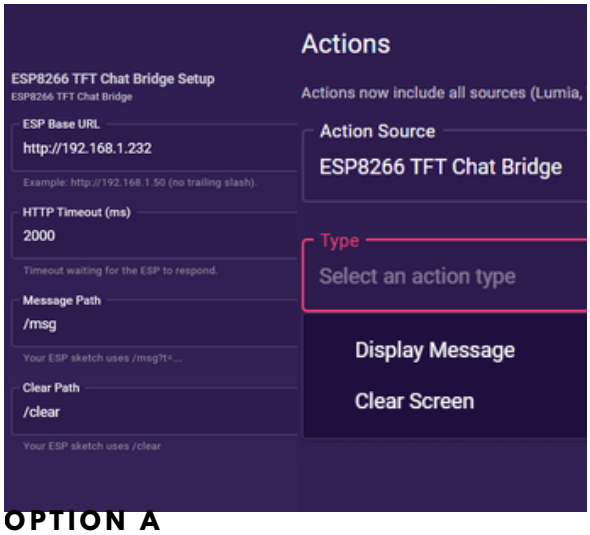
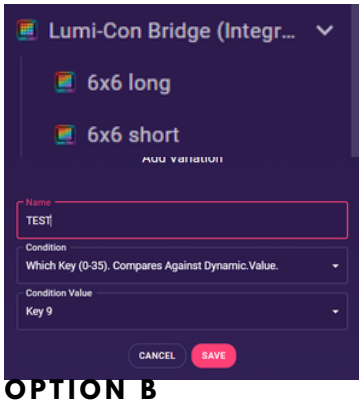
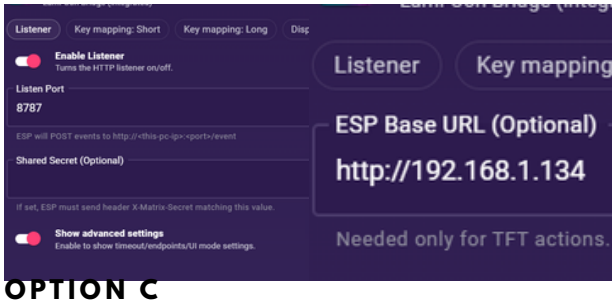
PLUGIN CONFIG



OPTION A



OPTION B



OPTION C

A) IF YOU CHOSE "JUST CHAT" (ESP8266-TFT-CHAT-BRIDGE)

1. OPEN THE PLUGIN'S SETTINGS IN LUMIA.
2. AFTER WI-FI CONNECTS, YOUR ESP8266 WILL SHOW ITS IP ADDRESS ON THE TFT.
3. ENTER THAT IP ADDRESS INTO THE PLUGIN'S "ESP BASE URL" / "DEVICE IP" FIELD (WHICHEVER YOUR PLUGIN SHOWS).
4. CLICK SAVE.
5. YOU SHOULD NOW SEE THE DISPLAY ACTIONS (FOR EXAMPLE: ADD LINE / SET STATUS / CLEAR), AND LUMIA SHOULD BE ABLE TO UPDATE THE SCREEN.

B) IF YOU CHOSE "JUST 6×6" (6X6 ISOLATED)

1. OPEN THE PLUGIN'S SETTINGS IN LUMIA.
2. CLICK SAVE.
3. YOU SHOULD NOW SEE THE 6×6 ALERTS APPEAR (TYPICALLY "6×6 SHORT" AND "6×6 LONG").
4. USE VARIATIONS (KEY 0–35) TO SET DIFFERENT ACTIONS PER BUTTON.

C) IF YOU CHOSE "BOTH" (LUMI-CON BRIDGE INTEGRATED V5.2)

1. OPEN THE INTEGRATED PLUGIN'S SETTINGS IN LUMIA.
2. MAKE SURE "ENABLE LISTENER" IS ON (THIS RECEIVES BUTTON EVENTS).
3. ENTER THE ESP8266 IP ADDRESS (SHOWN ON THE TFT AFTER WI-FI CONNECTS) INTO "ESP BASE URL" (THIS ENABLES SCREEN ACTIONS).
4. CLICK SAVE.
5. YOU SHOULD NOW SEE:
6. • ALERTS: 6×6 SHORT + 6×6 LONG (FOR BUTTONS)
7. • ACTIONS: DISPLAY ADD LINE / DISPLAY SET STATUS / DISPLAY CLEAR (FOR TFT)

FINAL CHECK (ALL BUILDS)

AFTER YOU CLICK SAVE:

• ALERTS AND/OR ACTIONS SHOULD APPEAR IN LUMIA FOR YOUR CHOSEN PLUGIN.

IF NOTHING APPEARS: DOUBLE CHECK YOU INSTALLED THE CORRECT PLUGIN, THEN REOPEN THE PLUGIN SETTINGS AND CLICK SAVE AGAIN.